

Limits

Limit Notation

To notate the number "L" that the y-values of a graph "f" approach as the x-values approach some number "a," we use the following notation.

$$\lim_{x \rightarrow a} f(x) = L$$

This expression is read as **the limit of f(x) as x approaches "a" is "L"**.

It can be understood simply as the y-values approach "L" as the x-values approach "a."

Finding Limits given Graphs

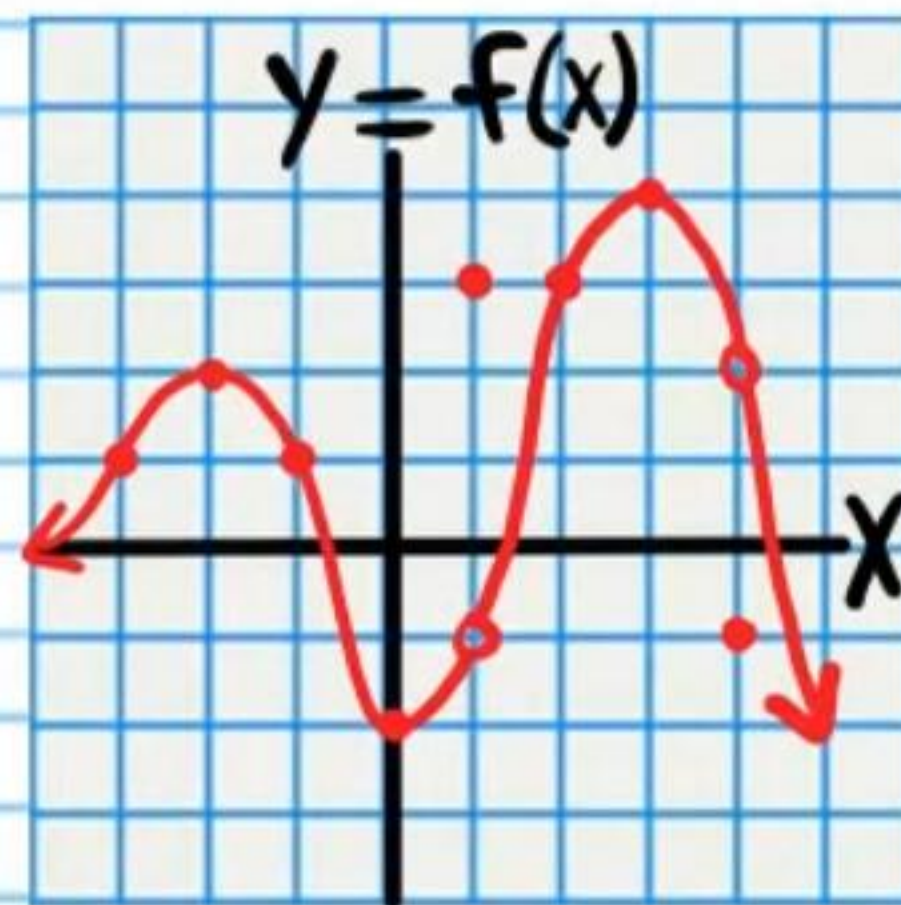
- Use the graph below to find the values of the expressions.

① $\lim_{x \rightarrow 2} f(x) = 3$ ⑤ $\lim_{x \rightarrow 1} f(x) = -1$

② $\lim_{x \rightarrow -3} f(x) = 1$ ⑥ $f(1) = 3$

③ $\lim_{x \rightarrow 4} f(x) = 2$ ⑦ $\lim_{x \rightarrow -2} f(x) = 2$

④ $f(4) = -1$ ⑧ $f(-2) = 2$



Continuous Function: A function whose graph has no breaks (i.e. holes, jumps, or vertical asymptotes).

Removable Discontinuity: A break in the graph of a function that can be repaired/removed by changing the value of **one** output of the function.

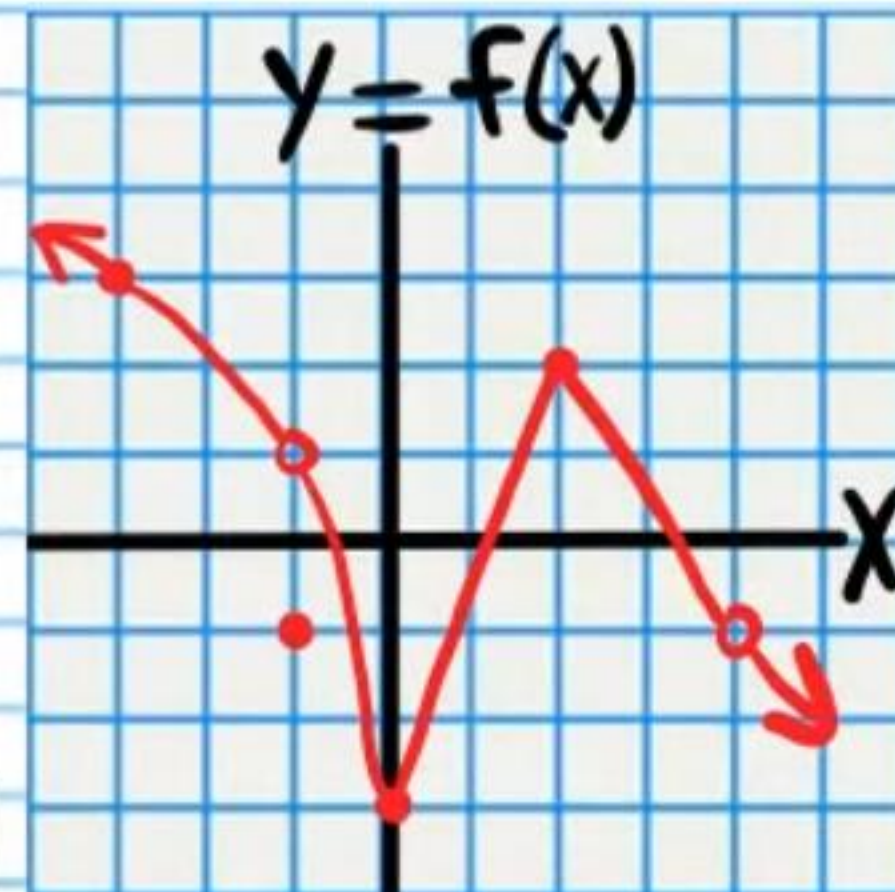
Use the graph below to find the values of the expressions.

⑨ $\lim_{x \rightarrow 4} f(x) = -1$ ⑬ $\lim_{x \rightarrow -1} f(x) = 1$

⑩ $f(4) = \text{DNE}$ ⑭ $f(-1) = -1$

⑪ $\lim_{x \rightarrow 0} f(x) = -3$ ⑮ $\lim_{x \rightarrow -3} f(x) = 3$

⑫ $f(0) = -3$ ⑯ $\lim_{x \rightarrow 2} f(x) = 2$



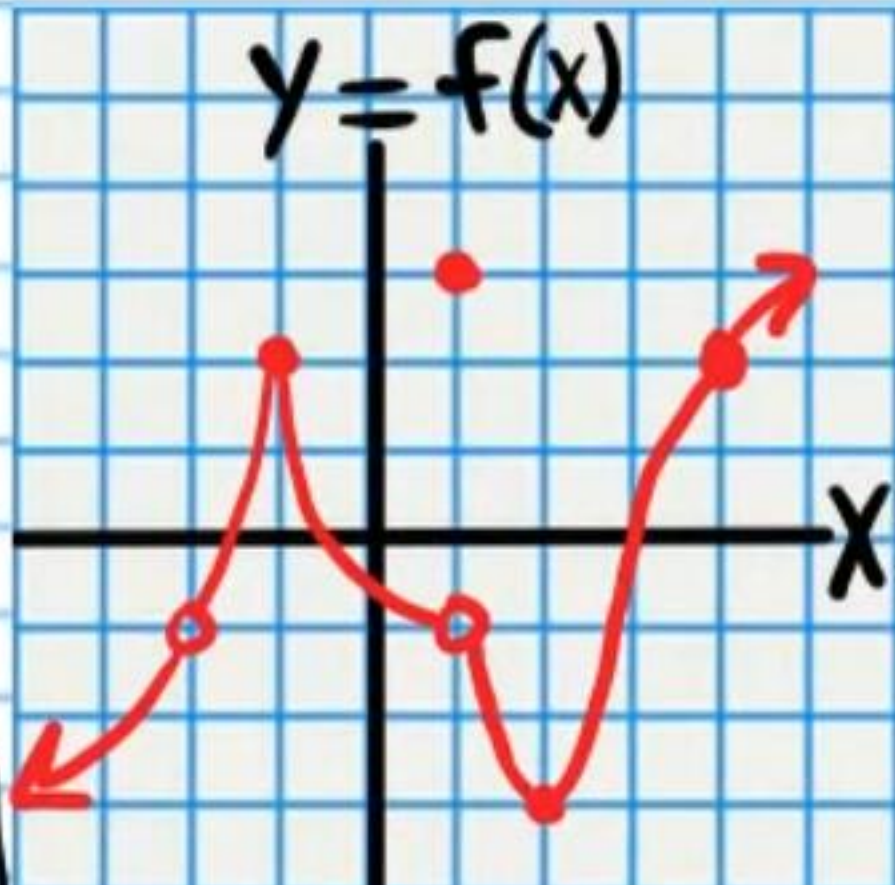
- Use the graph below to find the values of the expressions.

⑰ $\lim_{x \rightarrow -1} f(x) = 2$ ⑲ $\lim_{x \rightarrow 4} f(x) = 2$

⑱ $\lim_{x \rightarrow 2} f(x) = -3$ ⑳ $f(4) = 2$

㉑ $\lim_{x \rightarrow 1} f(x) = -1$ ㉓ $\lim_{x \rightarrow -2} f(x) = -1$

㉒ $f(1) = 3$ ㉔ $f(-2) = \text{DNE}$



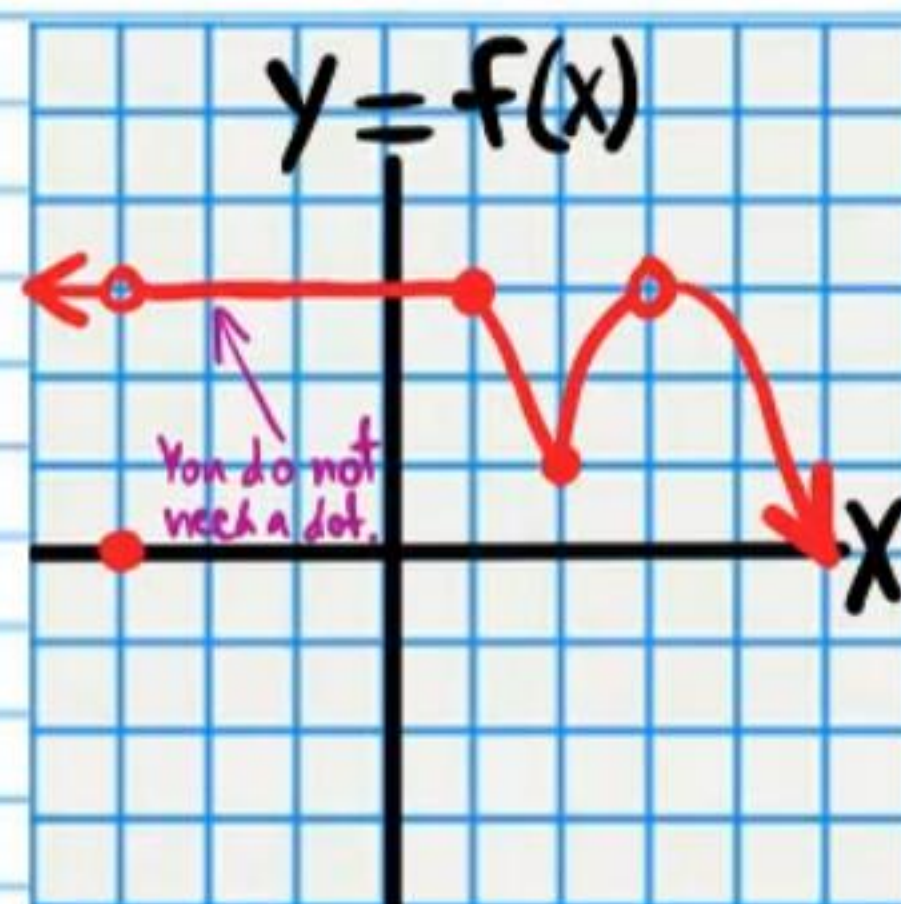
Use the graph below to find the values of the expressions.

25) $\lim_{x \rightarrow 3} f(x) = \boxed{3}$ 29) $\lim_{x \rightarrow -3} f(x) = \boxed{3}$

26) $f(3) = \boxed{\text{DNE}}$ 30) $f(-3) = \boxed{0}$

27) $\lim_{x \rightarrow 1} f(x) = \boxed{3}$ 31) $\lim_{x \rightarrow -2} f(x) = \boxed{3}$

28) $f(1) = \boxed{3}$ 32) $f(-2) = \boxed{3}$



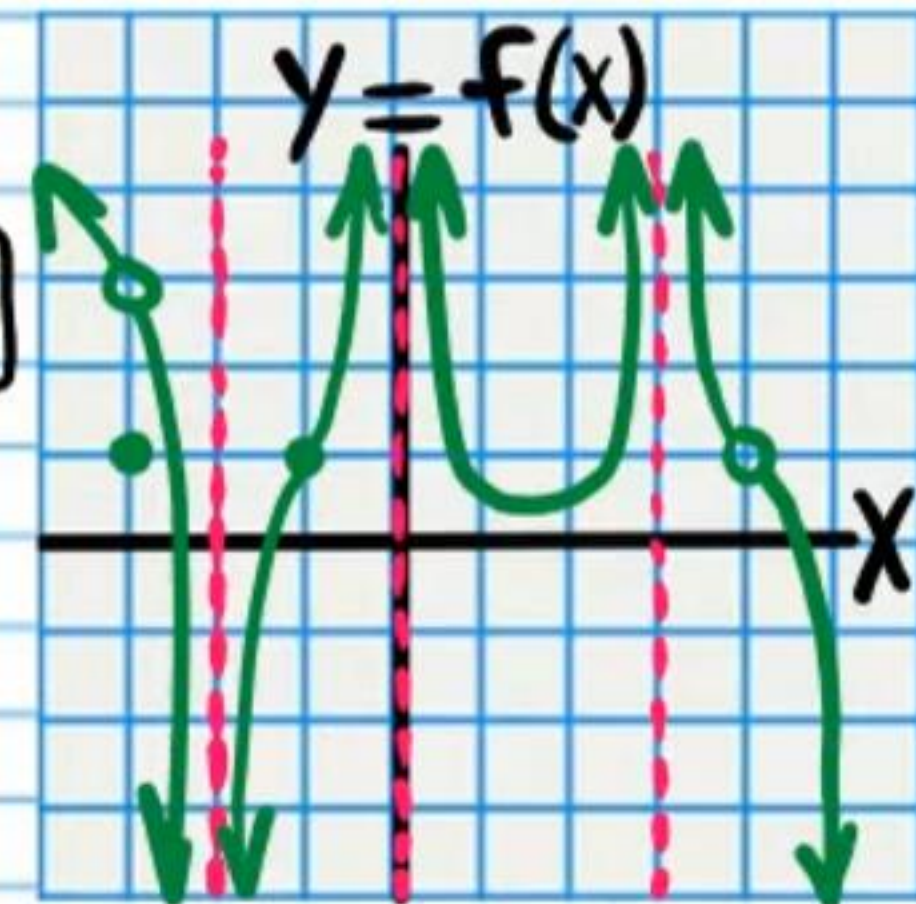
Use the graph below to find the values of the expressions.

41) $\lim_{x \rightarrow 0} f(x) = \boxed{\infty}$ 45) $\lim_{x \rightarrow 4} f(x) = \boxed{1}$

42) $\lim_{x \rightarrow -1} f(x) = \boxed{1}$ 46) $f(4) = \boxed{\text{DNE}}$

43) $f(-1) = \boxed{1}$ 47) $\lim_{x \rightarrow -3} f(x) = \boxed{3}$

44) $\lim_{x \rightarrow -2} f(x) = \boxed{-\infty}$ 48) $f(-3) = \boxed{1}$



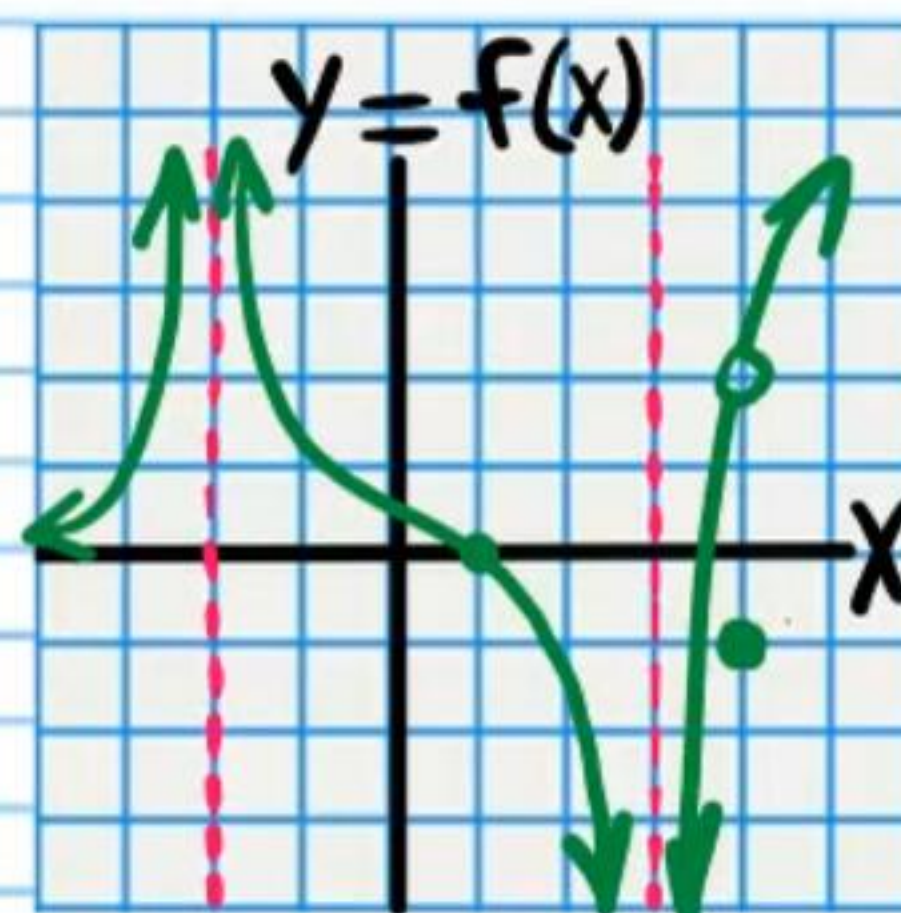
Use the graph below to find the values of the expressions.

33) $\lim_{x \rightarrow -2} f(x) = \boxed{\infty}$ 37) $\lim_{x \rightarrow 1} f(x) = \boxed{0}$

34) $\lim_{x \rightarrow 3} f(x) = \boxed{-\infty}$ 38) $f(1) = \boxed{0}$

35) $f(-2) = \boxed{\text{DNE}}$ 39) $\lim_{x \rightarrow 4} f(x) = \boxed{2}$

36) $f(3) = \boxed{\text{DNE}}$ 40) $f(4) = \boxed{-1}$



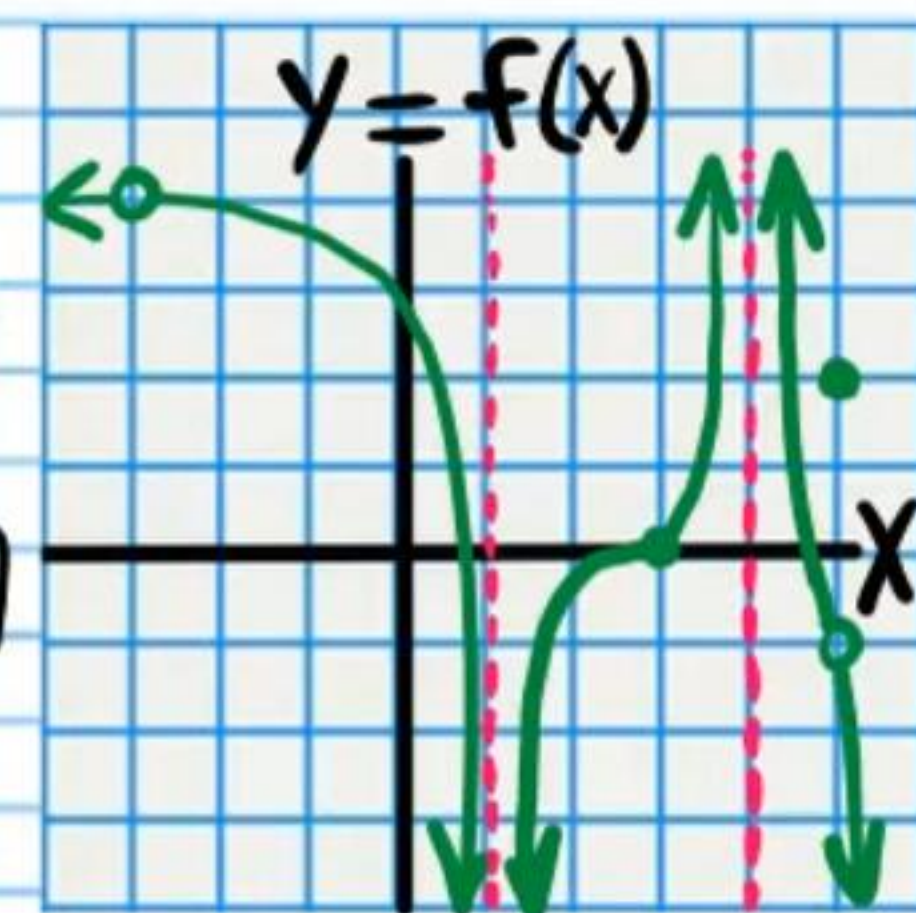
Use the graph below to find the values of the expressions.

49) $\lim_{x \rightarrow 1} f(x) = \boxed{-\infty}$ 53) $\lim_{x \rightarrow 4} f(x) = \boxed{\infty}$

50) $\lim_{x \rightarrow 3} f(x) = \boxed{0}$ 54) $f(1) = \boxed{\text{DNE}}$

51) $\lim_{x \rightarrow -3} f(x) = \boxed{4}$ 55) $\lim_{x \rightarrow 5} f(x) = \boxed{-1}$

52) $f(-3) = \boxed{\text{DNE}}$ 56) $f(5) = \boxed{2}$



Infinite Discontinuity: A break in the graph of a function where a vertical asymptote appears.

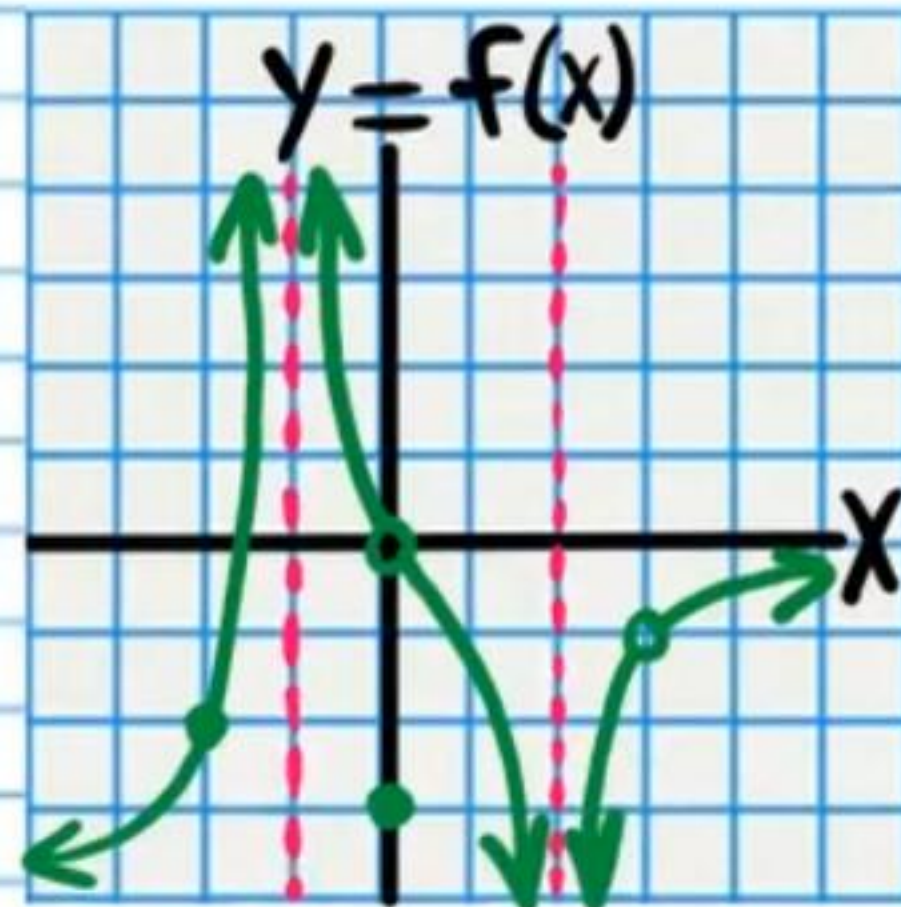
Use the graph below to find the values of the expressions.

57) $\lim_{x \rightarrow 2} f(x) = \boxed{-\infty}$ 61) $\lim_{x \rightarrow -2} f(x) = \boxed{-2}$

58) $\lim_{x \rightarrow -1} f(x) = \boxed{\infty}$ 62) $f(-2) = \boxed{-2}$

59) $\lim_{x \rightarrow 0} f(x) = \boxed{0}$ 63) $\lim_{x \rightarrow 3} f(x) = \boxed{-1}$

60) $f(0) = \boxed{-3}$ 64) $f(3) = \boxed{\text{DNE}}$



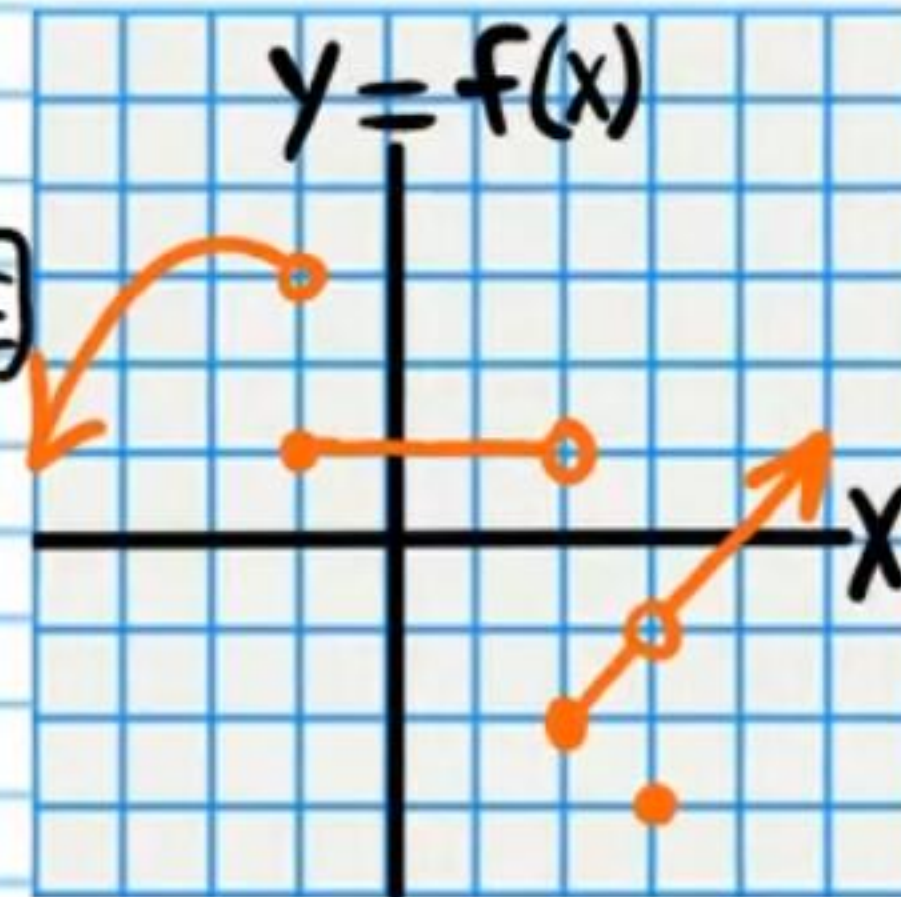
• Use the graph below to find the values of the expressions.

65) $\lim_{x \rightarrow 2^+} f(x) = \boxed{-2}$ 69) $\lim_{x \rightarrow -1^-} f(x) = \boxed{3}$

66) $\lim_{x \rightarrow 2^-} f(x) = \boxed{1}$ 70) $\lim_{x \rightarrow -1} f(x) = \boxed{\text{DNE}}$

67) $\lim_{x \rightarrow 2} f(x) = \boxed{\text{DNE}}$ 71) $\lim_{x \rightarrow 3} f(x) = \boxed{-1}$

68) $\lim_{x \rightarrow -1^+} f(x) = \boxed{1}$ 72) $f(3) = \boxed{-3}$



Jump Discontinuity: A break in the graph of a function where the values of the function seem to abruptly shift up or down to a different set of values.

$\lim_{x \rightarrow a} f(x) = L$ if and only if $\lim_{x \rightarrow a^+} f(x) = L$ and $\lim_{x \rightarrow a^-} f(x) = L$

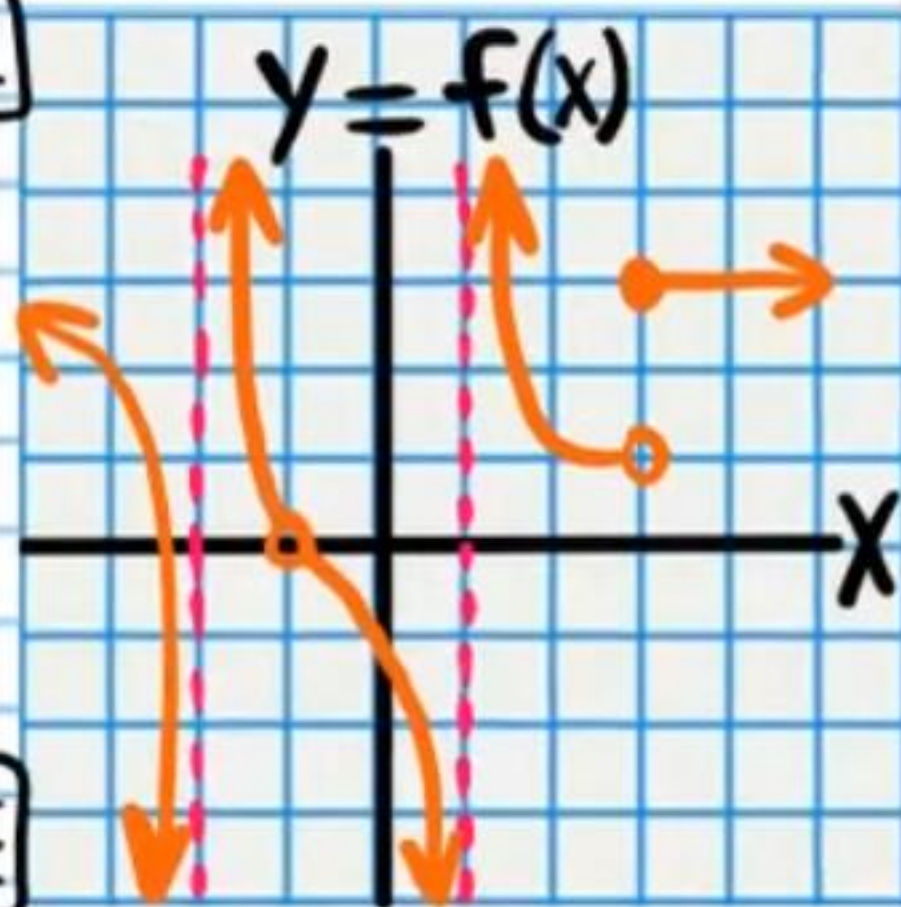
Use the graph below to find the values of the expressions.

73) $\lim_{x \rightarrow -2} f(x) = \boxed{\text{DNE}}$ 77) $\lim_{x \rightarrow 1} f(x) = \boxed{\text{DNE}}$

74) $\lim_{x \rightarrow -2^+} f(x) = \boxed{\infty}$ 78) $\lim_{x \rightarrow 1^-} f(x) = \boxed{-\infty}$

75) $\lim_{x \rightarrow -2^-} f(x) = \boxed{-\infty}$ 79) $\lim_{x \rightarrow -1} f(x) = \boxed{0}$

76) $\lim_{x \rightarrow 1^+} f(x) = \boxed{\infty}$ 80) $\lim_{x \rightarrow 3} f(x) = \boxed{\text{DNE}}$



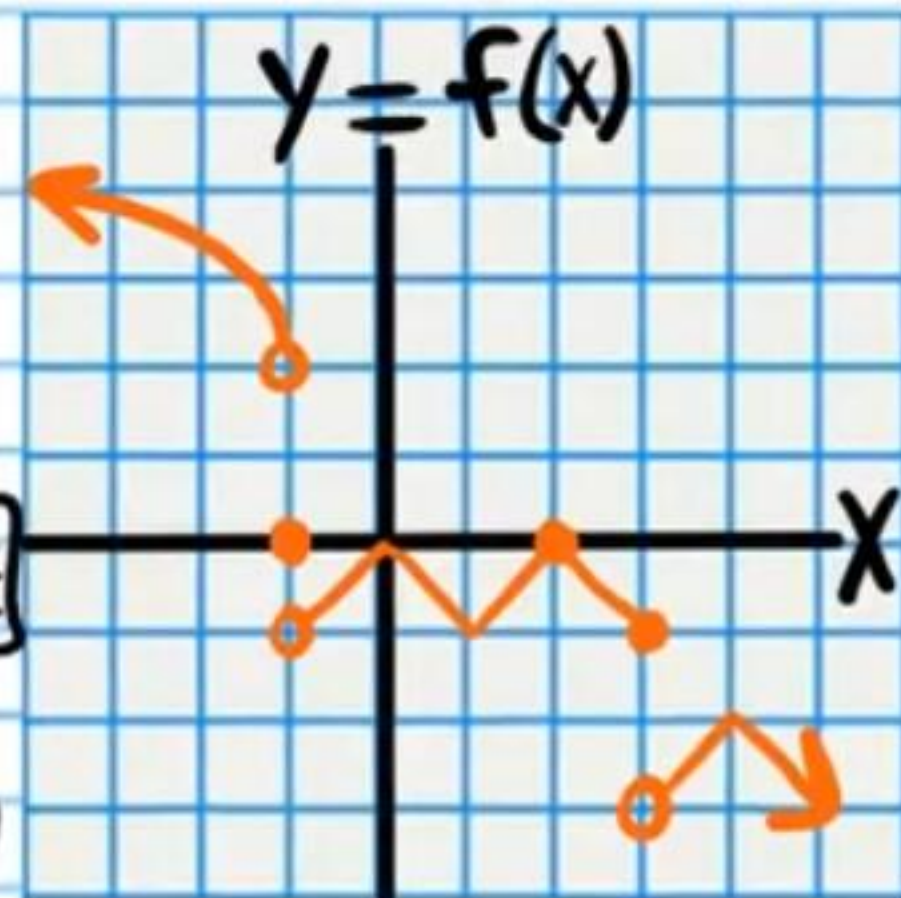
• Use the graph below to find the values of the expressions.

81) $\lim_{x \rightarrow -1^-} f(x) = \boxed{2}$ 85) $\lim_{x \rightarrow 3^+} f(x) = \boxed{-3}$

82) $\lim_{x \rightarrow -1^+} f(x) = \boxed{-1}$ 86) $\lim_{x \rightarrow 3^-} f(x) = \boxed{-1}$

83) $\lim_{x \rightarrow -1} f(x) = \boxed{\text{DNE}}$ 87) $\lim_{x \rightarrow 3} f(x) = \boxed{\text{DNE}}$

84) $f(-1) = \boxed{0}$ 88) $\lim_{x \rightarrow 2} f(x) = \boxed{0}$



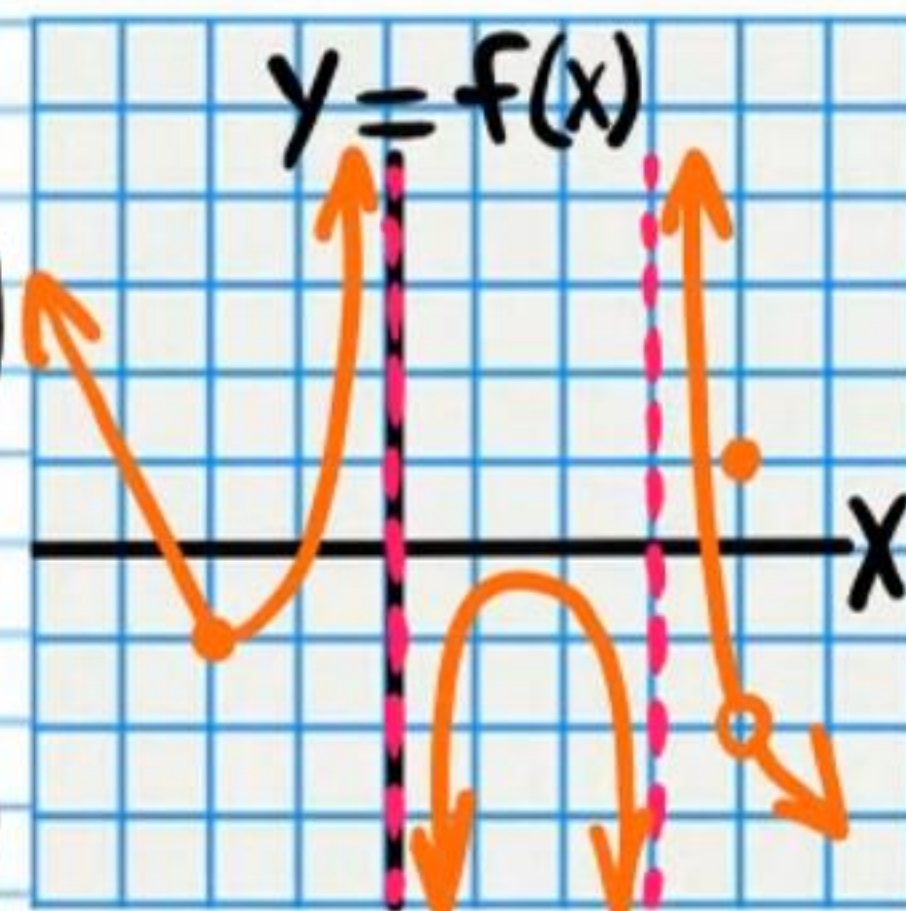
Use the graph below to find the values of the expressions.

89) $\lim_{x \rightarrow 0^+} f(x) = -\infty$ 93) $\lim_{x \rightarrow 3^+} f(x) = \infty$

90) $\lim_{x \rightarrow 0^-} f(x) = \infty$ 94) $\lim_{x \rightarrow 3} f(x) = \text{DNE}$

91) $\lim_{x \rightarrow 0} f(x) = \text{DNE}$ 95) $\lim_{x \rightarrow -2^+} f(x) = -1$

92) $\lim_{x \rightarrow 3^-} f(x) = -\infty$ 96) $\lim_{x \rightarrow -2} f(x) = -1$



Use the graph below to find the values of the expressions.

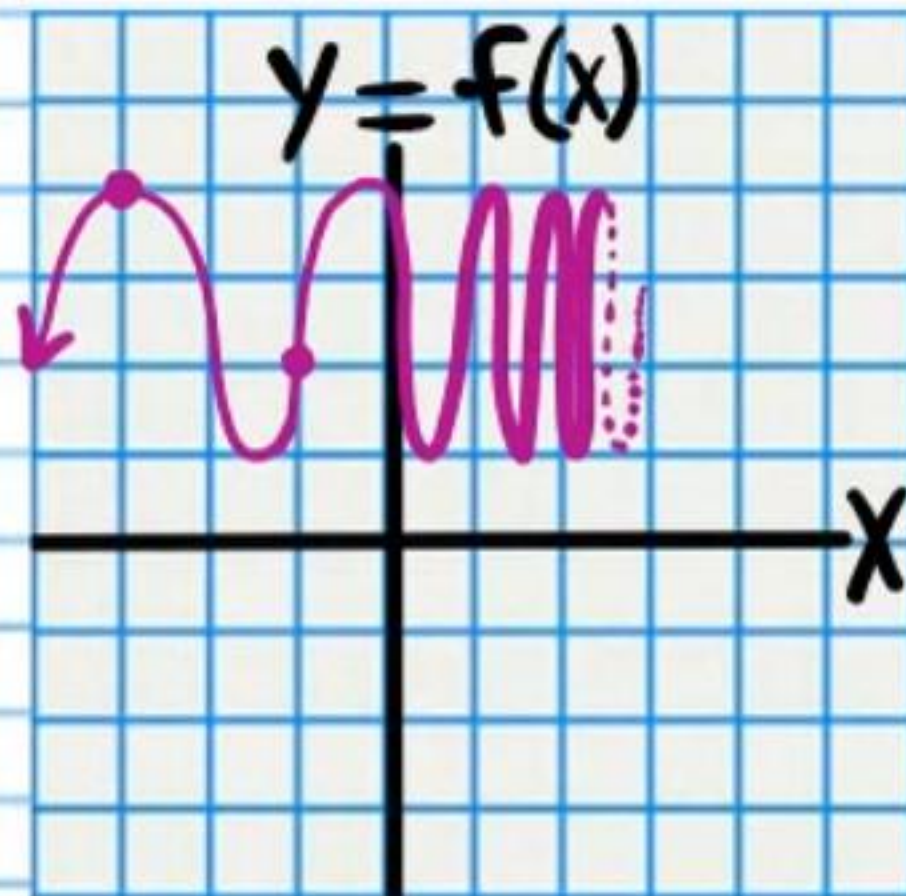
97) $\lim_{x \rightarrow 3^-} f(x) = \text{DNE}$ 102) $\lim_{x \rightarrow -1^+} f(x) = 2$

98) $\lim_{x \rightarrow -3^+} f(x) = 4$ 103) $\lim_{x \rightarrow -1^-} f(x) = 2$

99) $\lim_{x \rightarrow -3^-} f(x) = 4$ 104) $\lim_{x \rightarrow -1} f(x) = 2$

100) $\lim_{x \rightarrow -3} f(x) = 4$ 105) $f(-1) = 2$

101) $f(-3) = 4$



Use the graph below to find the values of the expressions.

106) $\lim_{x \rightarrow 0^-} f(x) = \text{DNE}$ 112) $f(4) = 1$

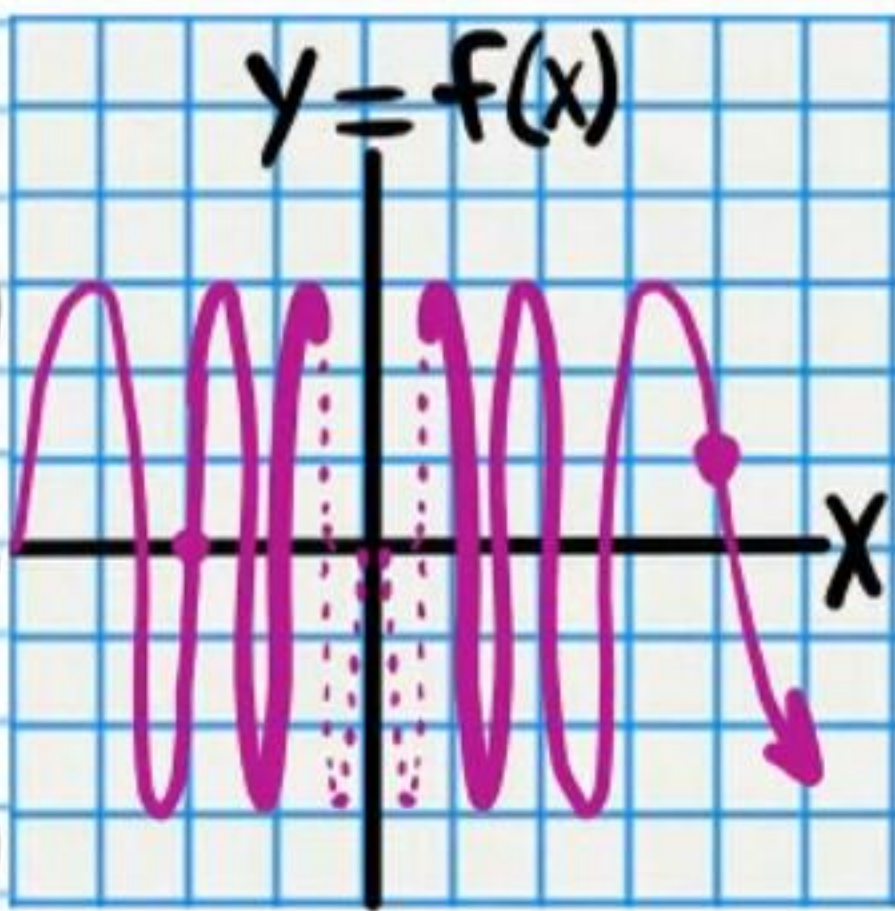
107) $\lim_{x \rightarrow 0^+} f(x) = \text{DNE}$ 113) $\lim_{x \rightarrow -2^+} f(x) = 0$

108) $\lim_{x \rightarrow 0} f(x) = \text{DNE}$ 114) $\lim_{x \rightarrow -2^-} f(x) = 0$

109) $\lim_{x \rightarrow 4^-} f(x) = 1$ 115) $\lim_{x \rightarrow -2} f(x) = 0$

110) $\lim_{x \rightarrow 4^+} f(x) = 1$ 116) $f(-2) = 0$

111) $\lim_{x \rightarrow 4} f(x) = 1$



The Definition of Continuity at Some X-value: A function is continuous at $x = a$ if $\lim_{x \rightarrow a} f(x) = f(a)$.

In order for $\lim_{x \rightarrow a} f(x) = f(a)$, three things must be true.

i) $f(a)$ must exist.

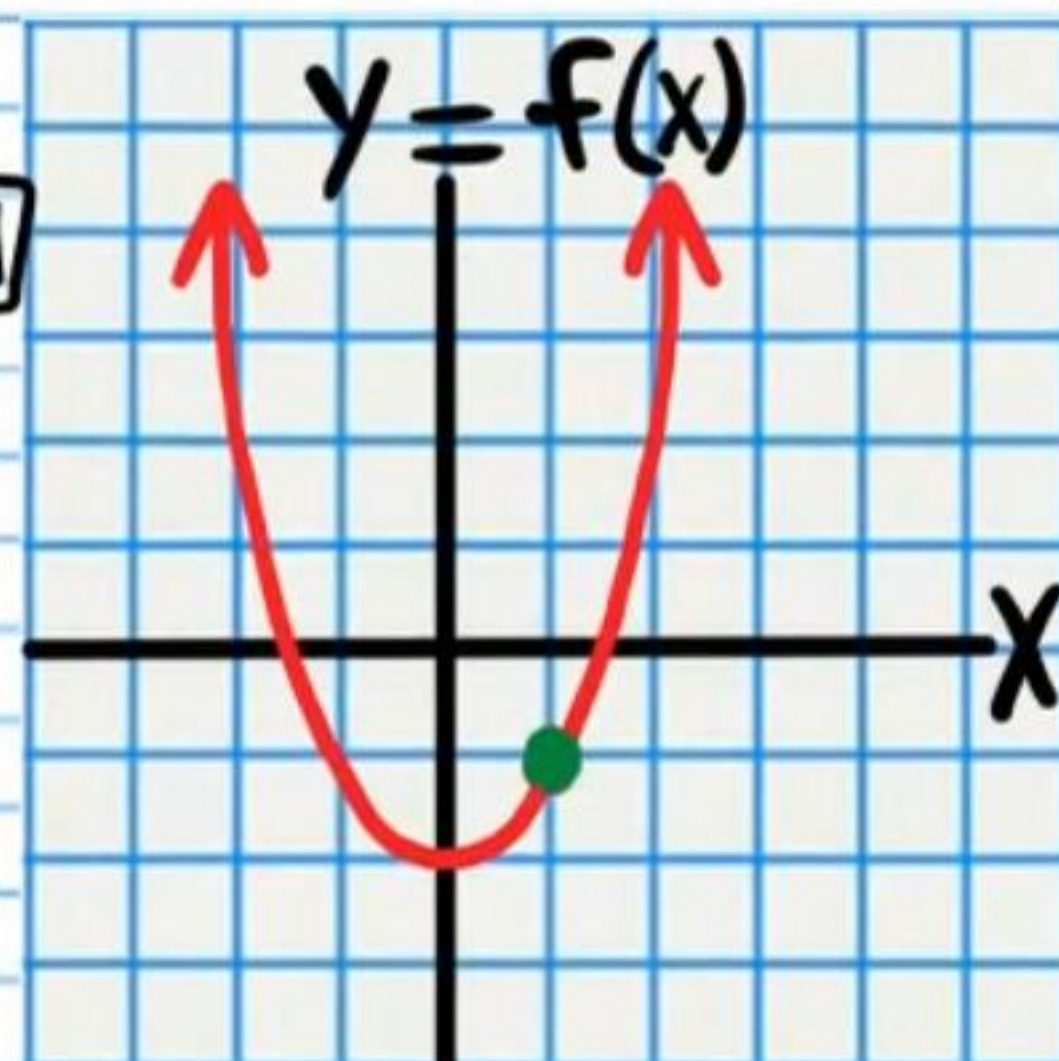
ii) $\lim_{x \rightarrow a} f(x)$ must exist (implying that $\lim_{x \rightarrow a^+} f(x)$ and $\lim_{x \rightarrow a^-} f(x)$ both exist and are equal).

iii) $f(a)$ and $\lim_{x \rightarrow a} f(x)$ must be equal.

Finding Limits given Equations

Example I)

$$\lim_{x \rightarrow 1} (x^2 - 2) = 1^2 - 2 = 1 - 2 = \boxed{-1}$$

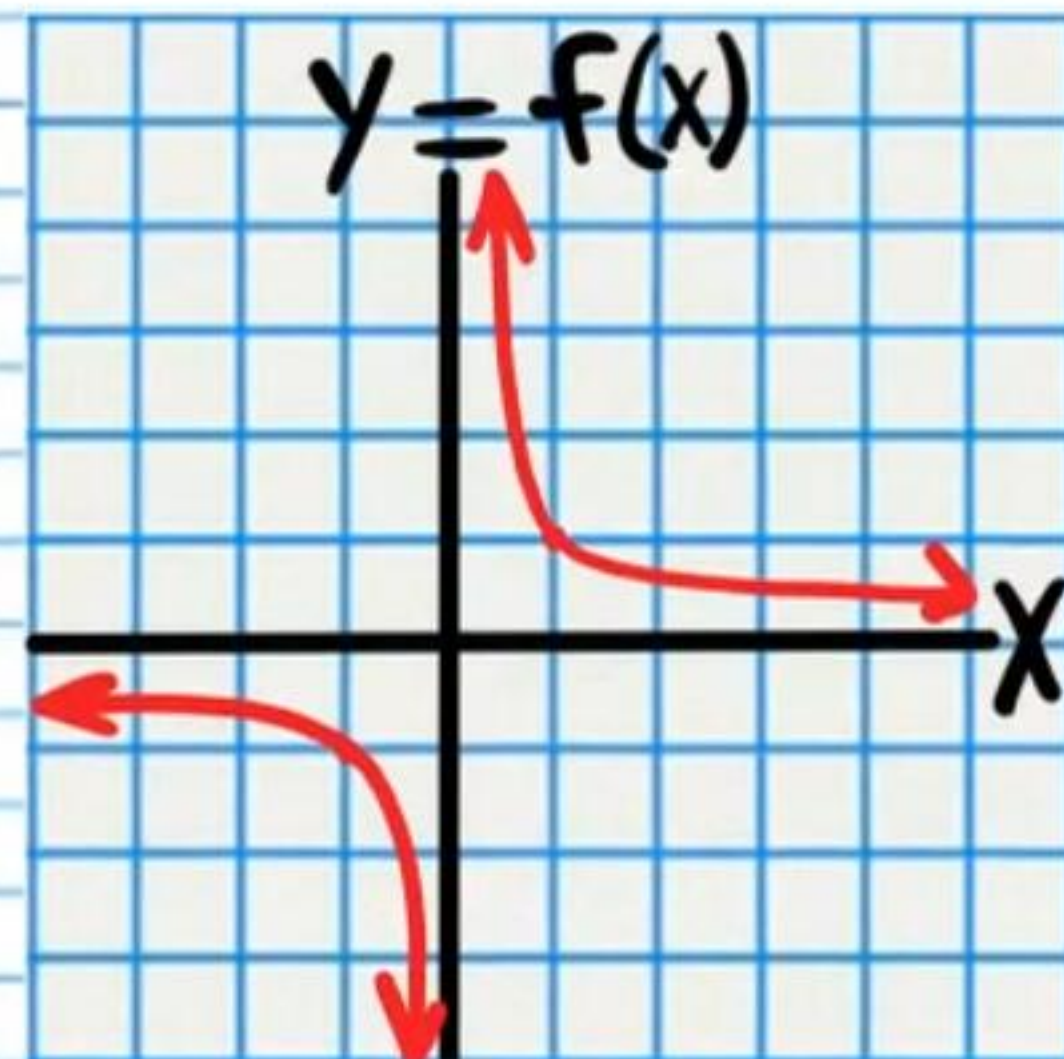


Example III)

$$i) \lim_{x \rightarrow 0^+} \frac{1}{x} = \boxed{\infty}$$

$$ii) \lim_{x \rightarrow 0^-} \frac{1}{x} = \boxed{-\infty}$$

$$iii) \lim_{x \rightarrow 0} \frac{1}{x} = \boxed{DNE}$$



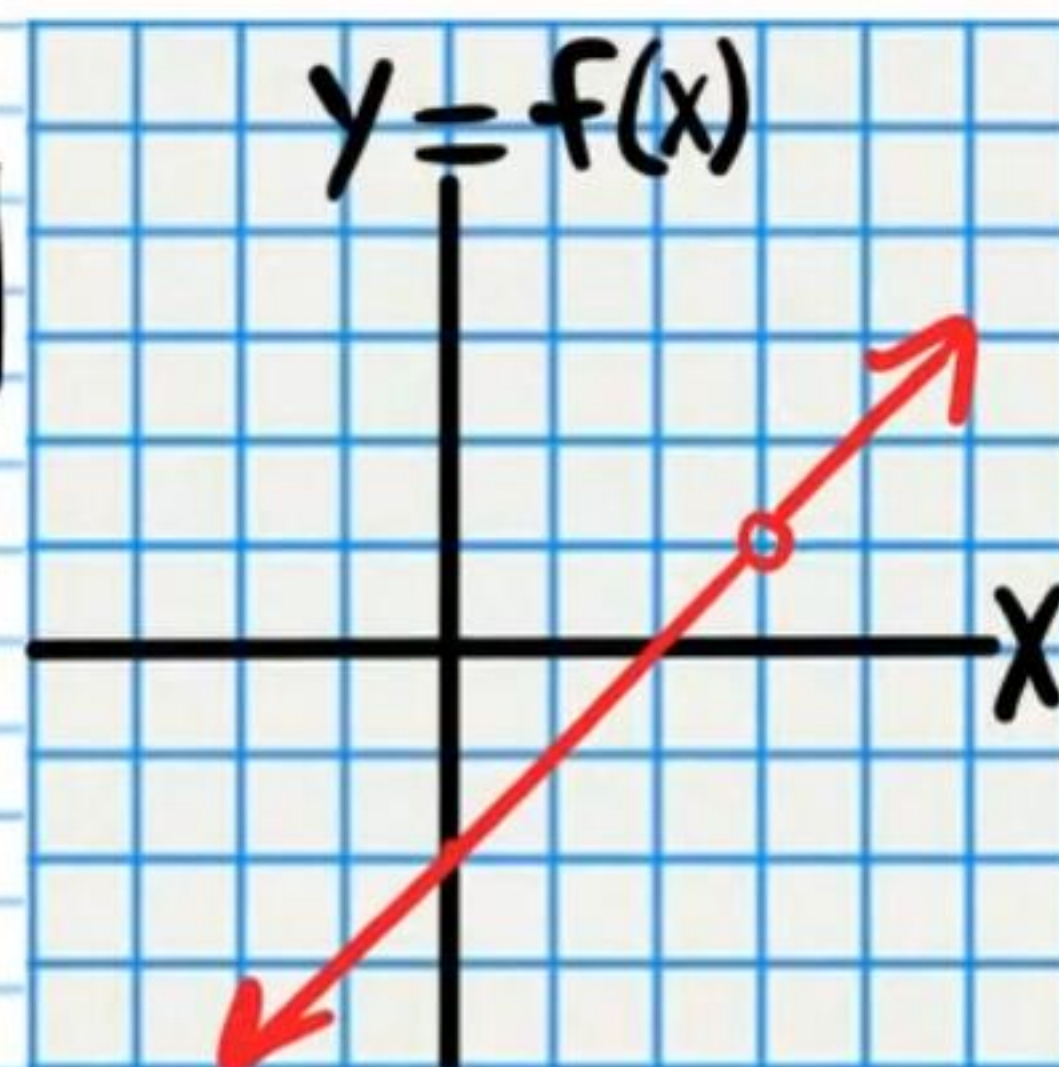
Example II)

$$\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{x - 3} = \lim_{x \rightarrow 3} \left[\frac{(x-2)(x-3)}{x-3} \right]$$

$$= \lim_{x \rightarrow 3} (x-2)$$

$$= 3 - 2$$

$$= \boxed{1}$$



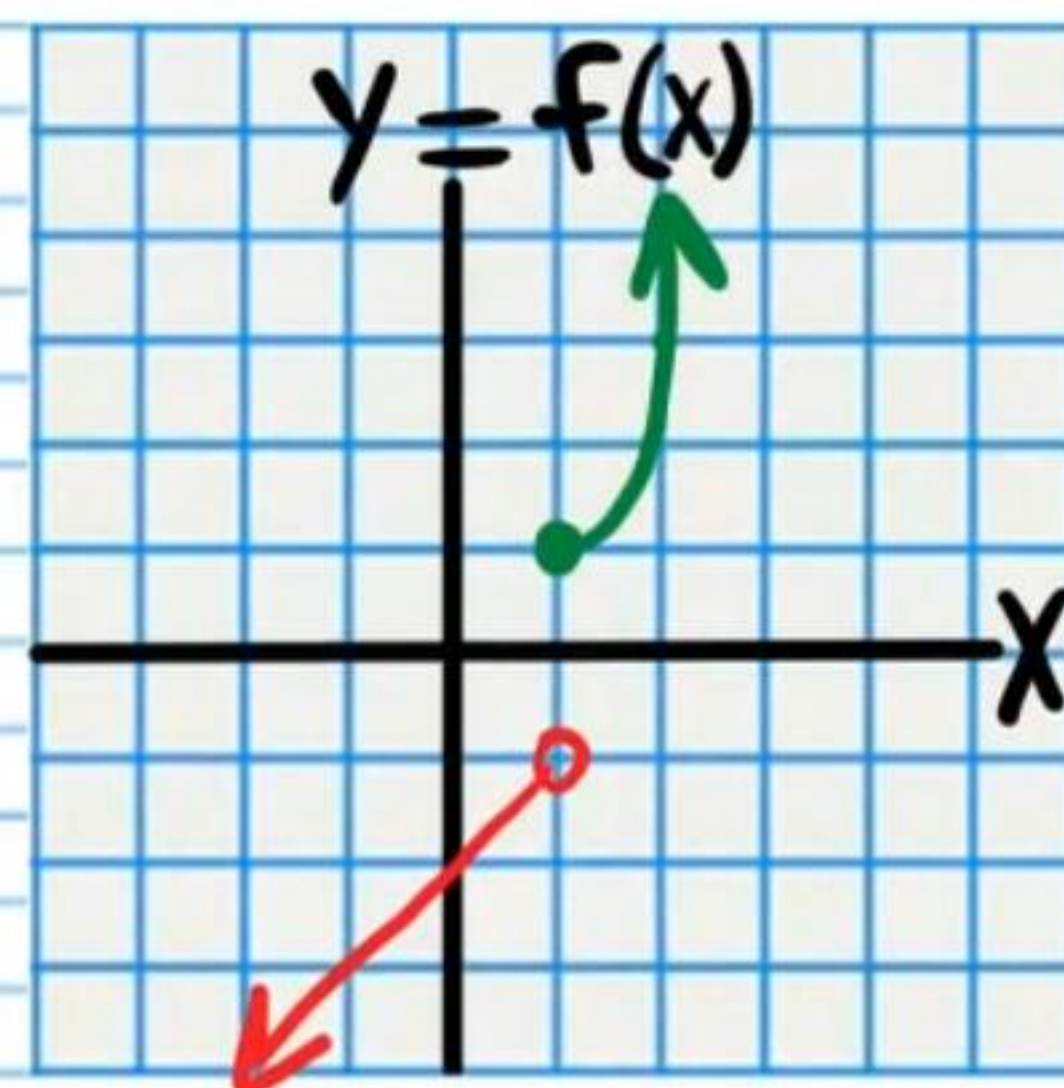
Example IV)

$$f(x) = \begin{cases} x-2, & x < 1 \\ 3^{x-1}, & x \geq 1 \end{cases}$$

$$i) \lim_{x \rightarrow 1^+} f(x) = \boxed{1}$$

$$ii) \lim_{x \rightarrow 1^-} f(x) = \boxed{-1}$$

$$iii) \lim_{x \rightarrow 1} f(x) = \boxed{DNE}$$



Finding Limits at X-Values where the Function Is Continuous

Find the value of each expression below.

$$\bullet \textcircled{117} \lim_{x \rightarrow -2} (x^3 - 2x + 8) = (-2)^3 - 2(-2) + 8 = -8 + 4 + 8 = \boxed{4}$$

$$\textcircled{118} \lim_{x \rightarrow 7} \frac{1}{x} = \boxed{\frac{1}{7}}$$

$$\textcircled{119} \lim_{x \rightarrow 0} 3^x = 3^0 = \boxed{1}$$

$$\textcircled{120} \lim_{x \rightarrow 4} \log_2 x = \log_2 4 = \log_2 2^2 = \boxed{2}$$

$$\textcircled{121} \lim_{x \rightarrow -1} \sqrt{2x+3} = \sqrt{2(-1)+3} = \sqrt{-2+3} = \sqrt{1} = \boxed{1}$$

$$\bullet \textcircled{122} \lim_{x \rightarrow 10} x^3 = 10^3 = \boxed{1000}$$

$$\textcircled{123} \lim_{x \rightarrow -4} (x^2 + 3x - 7) = (-4)^2 + 3(-4) - 7 = 16 - 12 - 7 = 4 - 7 = \boxed{-3}$$

$$\textcircled{124} \lim_{x \rightarrow 0} \ln(4x+10) = \ln(4(0)+10) = \ln(0+10) = \boxed{\ln 10}$$

$$\textcircled{125} \lim_{x \rightarrow 8} |3x| = |3(8)| = |24| = \boxed{24}$$

$$\textcircled{126} \lim_{x \rightarrow -3} \log_4(1-x) = \log_4(1-(-3)) = \log_4(1+3) = \log_4 4^1 = \boxed{1}$$

$$\textcircled{127} \lim_{x \rightarrow 0} \frac{2}{1-x} = \frac{2}{1-0} = \frac{2}{1} = \boxed{2}$$

You may
use your
calculator
in
the form of
logarithm.

Finding Limits at Removable Discontinuities

$$\bullet \textcircled{128} \lim_{x \rightarrow -6} \frac{x+6}{x^2+7x+6} = \lim_{x \rightarrow -6} \frac{\overset{\text{Factoring}}{\cancel{x+6}}}{\underset{\text{Factoring}}{(x+1)\cancel{(x+6)}}} = \lim_{x \rightarrow -6} \frac{1}{x+1} = \frac{1}{-6+1} = \frac{1}{-5} = \boxed{-\frac{1}{5}}$$

$$\textcircled{129} \lim_{x \rightarrow 0} \frac{20x^3 - 15x^2}{8x^2} = \lim_{x \rightarrow 0} \frac{\overset{\text{Factoring}}{5x^2(4x-3)}}{\underset{\text{Factoring}}{8x^2}} = \lim_{x \rightarrow 0} \left[\frac{5(4x-3)}{8} \right]$$

$$= \frac{5(4(0)-3)}{8}$$

$$= \boxed{-\frac{15}{8}}$$

$$\textcircled{130} \lim_{x \rightarrow \frac{5}{6}} \frac{36x^2 - 25}{5 - 6x} = \lim_{x \rightarrow \frac{5}{6}} \left[\frac{(6x-5)(6x+5)}{5-6x} \right] = \lim_{x \rightarrow \frac{5}{6}} \left[\frac{(6x-5)(6x+5)}{-(-5+6x)} \right]$$

$$= \lim_{x \rightarrow \frac{5}{6}} \frac{6x+5}{-1}$$

$$= \frac{6\left(\frac{5}{6}\right) + 5}{-1}$$

$$= \frac{5+5}{-1}$$

$$= \boxed{-10}$$

$$\bullet \textcircled{131} \lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{x - 1} = \lim_{x \rightarrow 1} \left[\frac{(x+4)(x-1)}{x-1} \right] = \lim_{x \rightarrow 1} (x+4)$$

$$= 1+4$$

$$= \boxed{5}$$

$$\textcircled{132} \lim_{x \rightarrow 0} \frac{3x^4}{9x^6 - 5x^4} = \lim_{x \rightarrow 0} \left[\frac{3x^4}{x^4(9x^2 - 5)} \right] = \lim_{x \rightarrow 0} \frac{3}{9x^2 - 5}$$

$$= \frac{3}{9(0)^2 - 5}$$

$$= \frac{3}{0-5}$$

$$= \boxed{-\frac{3}{5}}$$

$$\textcircled{133} \lim_{x \rightarrow -3} \frac{x+3}{x^2-9} = \lim_{x \rightarrow -3} \frac{x+3}{(x+3)(x-3)} = \lim_{x \rightarrow -3} \frac{1}{x-3} = \frac{1}{-3-3}$$

$$= \boxed{-\frac{1}{6}}$$

Expanding

$$\begin{aligned} \bullet \textcircled{134} \lim_{x \rightarrow 0} \frac{(2+x)^2 - 4}{x} &= \lim_{x \rightarrow 0} \frac{(2+x)(2+x) - 4}{x} \\ &= \lim_{x \rightarrow 0} \frac{4 + 2x + 2x + x^2 - 4}{x} \\ &= \lim_{x \rightarrow 0} \frac{4x + x^2}{x} \\ &= \lim_{x \rightarrow 0} \frac{x(4+x)}{x} \\ &= \lim_{x \rightarrow 0} (4+x) \\ &= 4+0 \\ &= \boxed{4} \end{aligned}$$

$$\textcircled{135} \lim_{x \rightarrow 1} \frac{(x-2)^2 - 1}{x-1} = \lim_{x \rightarrow 1} \frac{x^2 - 4x + 4 - 1}{x-1}$$

$$= \lim_{x \rightarrow 1} \frac{x^2 - 4x + 3}{x-1}$$

$$= \lim_{x \rightarrow 1} \left[\frac{(x-3)(x-1)}{x-1} \right]$$

$$= \lim_{x \rightarrow 1} (x-3)$$

$$= 1-3$$

$$= \boxed{-2}$$

$$(136) \lim_{x \rightarrow 10} \frac{-10+x}{181-(x-9)^2-18x} = \lim_{x \rightarrow 10} \frac{-10+x}{181-(x^2-18x+81)-18x}$$

$$= \lim_{x \rightarrow 10} \frac{-10+x}{181-x^2+18x-81-18x}$$

$$= \lim_{x \rightarrow 10} \frac{-10+x}{100-x^2}$$

$$= \lim_{x \rightarrow 10} \frac{-10+x}{(10-x)(10+x)}$$

$$= \lim_{x \rightarrow 10} \frac{-10+x}{-(-10+x)(10+x)}$$

$$= \lim_{x \rightarrow 10} \frac{-1}{10+x}$$

$$= \frac{-1}{10+10}$$

$$= \boxed{-\frac{1}{20}}$$

$$(137) \lim_{x \rightarrow 0} \frac{x}{(x+7)^2-49} = \lim_{x \rightarrow 0} \frac{x}{x^2+14x+49-49}$$

$$= \lim_{x \rightarrow 0} \frac{x}{x^2+14x}$$

$$= \lim_{x \rightarrow 0} \frac{x}{x(x+14)}$$

$$= \lim_{x \rightarrow 0} \frac{1}{x+14}$$

$$= \frac{1}{0+14}$$

$$= \boxed{\frac{1}{14}}$$

$$\textcircled{138} \lim_{x \rightarrow 9} \frac{25 - (-4+x)^2}{x-9} = \lim_{x \rightarrow 9} \frac{25 - (16 - 8x + x^2)}{x-9}$$

$$= \lim_{x \rightarrow 9} \frac{25 - 16 + 8x - x^2}{x-9}$$

$$= \lim_{x \rightarrow 9} \frac{9 + 8x - x^2}{x-9}$$

$$= \lim_{x \rightarrow 9} \frac{-x^2 + 8x + 9}{x-9}$$

$$= \lim_{x \rightarrow 9} \frac{-(x^2 - 8x - 9)}{x-9}$$

$$= \lim_{x \rightarrow 9} \left[- \frac{(x-9)(x+1)}{x-9} \right]$$

$$= \lim_{x \rightarrow 9} [-(x+1)]$$

$$= -(9+1)$$

$$= \boxed{-10}$$

$$\textcircled{139} \lim_{x \rightarrow 1} \frac{x-1}{(x+1)^2 - 2x - 2} = \lim_{x \rightarrow 1} \frac{x-1}{x^2 + 2x + 1 - 2x - 2}$$

$$= \lim_{x \rightarrow 1} \frac{x-1}{x^2 - 1}$$

$$= \lim_{x \rightarrow 1} \left[\frac{x-1}{(x-1)(x+1)} \right]$$

$$= \lim_{x \rightarrow 1} \frac{1}{x+1}$$

$$= \frac{1}{1+1}$$

$$= \boxed{\frac{1}{2}}$$